

Advanced Optimization

Chapter 6 — Quasi-Newton Methods

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Reading: Nocedal & Wright, *Numerical Optimization*, 2nd ed., Chapter 6

A crashing computer, and one of the most creative ideas in the field

Mid-1950s, Argonne: W. C. Davidon is running a long coordinate-descent calculation. The machine keeps crashing before it finishes. He looks for a way to *accelerate* the iteration — and invents the first quasi-Newton method.

Fletcher and Powell show it is far faster and more reliable than anything else; nonlinear optimization is transformed “overnight”. Davidon’s paper is *rejected*; it circulates as a technical report for thirty years and finally appears in the first issue of *SIAM J. Optimization*, 1991.

The idea

Like steepest descent, require **only gradients**. But *measure the change in gradients* between iterates and use it to build a model of f good enough for **superlinear** convergence.

- ▶ Dramatic improvement over steepest descent, especially on hard problems.
- ▶ Often *more* efficient than Newton, since no second derivatives and no linear solve.
- ▶ Automatic differentiation (Ch. 8) has made Newton’s method easier to use — but second derivatives via AD can still be costly or unavailable. Quasi-Newton remains everywhere.

This chapter: small and medium n . Large-scale variants (L-BFGS): Chapter 7.

Setting: Newton's method with a stand-in Hessian

Quadratic model at x_k ,

$$m_k(p) = f_k + \nabla f_k^T p + \frac{1}{2} p^T B_k p, \quad (6.1)$$

B_k symmetric positive definite, to be *updated* each iteration. Its minimizer is the search direction:

$$p_k = -B_k^{-1} \nabla f_k, \quad x_{k+1} = x_k + \alpha_k p_k, \quad (6.2-6.3)$$

with α_k satisfying the Wolfe conditions (3.6).

This is line-search Newton with B_k in place of $\nabla^2 f_k$. The whole question is:

How should B_{k+1} be built from B_k and what we learned on the last step?

Davidon's answer: don't recompute — *update*, using the curvature measured along the step.

The secant equation

Require the new model $m_{k+1}(p) = f_{k+1} + \nabla f_{k+1}^T p + \frac{1}{2} p^T B_{k+1} p$ to match the gradient of f at *both* of the last two iterates.

At x_{k+1} ($p = 0$): $\nabla m_{k+1}(0) = \nabla f_{k+1}$ — automatic. At x_k ($p = -\alpha_k p_k$):

$$\nabla m_{k+1}(-\alpha_k p_k) = \nabla f_{k+1} - \alpha_k B_{k+1} p_k \stackrel{!}{=} \nabla f_k \quad \implies \quad B_{k+1} \alpha_k p_k = \nabla f_{k+1} - \nabla f_k. \quad (6.4)$$

Notation, and the secant equation

$$s_k = x_{k+1} - x_k = \alpha_k p_k, \quad y_k = \nabla f_{k+1} - \nabla f_k, \quad (6.5)$$

$$\boxed{B_{k+1} s_k = y_k} \quad (6.6)$$

Compare Ch. 2, (2.16): for the true Hessian, $\nabla^2 f_k s_k \approx y_k$ up to $o(\|s_k\|)$. We ask the approximation to satisfy this *exactly*, along the one direction we just traveled. One equation; $n(n+1)/2$ unknowns.

The curvature condition — and why the Wolfe line search was never optional

Premultiply (6.6) by s_k^T : a positive definite B_{k+1} can map s_k to y_k only if

$$s_k^T y_k > 0. \quad (6.7)$$

- ▶ f strongly convex: (6.7) holds for *any* pair of points (Exercise 6.1).
- ▶ f nonconvex: it can fail. Then the line search must enforce it.

The Wolfe curvature condition delivers exactly (6.7)

From (3.6b), $\nabla f_{k+1}^T s_k \geq c_2 \nabla f_k^T s_k$, hence

$$y_k^T s_k \geq (c_2 - 1) \alpha_k \nabla f_k^T p_k > 0, \quad (6.8)$$

since $c_2 < 1$ and p_k is a descent direction. Strong Wolfe works too.

This is the promise made in Chapter 3 finally paid: the second Wolfe condition was introduced there to rule out short steps; here we see its other job. *Backtracking on Armijo alone does not guarantee (6.7).*

Pinning down B_{k+1} : closest to B_k in the right norm

Under (6.7) the secant equation has infinitely many symmetric positive definite solutions. Choose the one *closest to what we already have*:

$$\min_B \|B - B_k\| \quad \text{s.t.} \quad B = B^T, \quad Bs_k = y_k. \quad (6.9)$$

Different norms give different quasi-Newton methods.

Weighted Frobenius norm

$$\|A\|_W = \|W^{1/2}AW^{1/2}\|_F, \quad W \text{ any matrix with } Wy_k = s_k. \quad (6.10)$$

For concreteness $W = \bar{G}_k^{-1}$ with the **average Hessian**

$$\bar{G}_k = \int_0^1 \nabla^2 f(x_k + \tau \alpha_k p_k) d\tau, \quad y_k = \bar{G}_k s_k \quad (\text{Taylor, integral form}). \quad (6.11\text{--}6.12)$$

Why this norm: it is *non-dimensional*, so the update is **invariant to rescaling of the variables**. We do not want the method to depend on whether x is in meters or millimeters. (W never needs to be computed — it drops out of the answer.)

DFP: the original update

The unique solution of (6.9) with the norm (6.10):

$$\textbf{(DFP)} \quad B_{k+1} = (I - \rho_k y_k s_k^T) B_k (I - \rho_k s_k y_k^T) + \rho_k y_k y_k^T, \quad \rho_k = \frac{1}{y_k^T s_k}. \quad (6.13\text{--}6.14)$$

Davidon 1959; studied, implemented, popularized by Fletcher and Powell.

For the iteration we want the *inverse* $H_k = B_k^{-1}$ (so $p_k = -H_k \nabla f_k$ is a matrix–vector product). Sherman–Morrison–Woodbury gives

$$\textbf{(DFP)} \quad H_{k+1} = H_k - \frac{H_k y_k y_k^T H_k}{y_k^T H_k y_k} + \frac{s_k s_k^T}{y_k^T s_k}. \quad (6.15)$$

The fundamental idea of quasi-Newton updating

(6.13) and (6.15) are **rank-two** modifications. Never rebuild the Hessian; apply a cheap correction that folds in the most recent curvature information.

DFP is effective — but was soon superseded.

BFGS: the same argument, applied to the inverse

Impose the conditions on H_{k+1} instead of B_{k+1} : symmetric, positive definite, and the secant equation in the form $H_{k+1}y_k = s_k$. Closest to H_k :

$$\min_H \|H - H_k\| \quad \text{s.t.} \quad H = H^T, \quad Hy_k = s_k, \quad (6.16)$$

same weighted norm, now with $Ws_k = y_k$ (e.g. $W = \bar{G}_k$). Unique solution:

$$\text{(BFGS)} \quad \boxed{H_{k+1} = (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T} \quad (6.17)$$

Broyden, Fletcher, Goldfarb, Shanno — independently, 1970.

Corresponding update of B_k (again via Sherman–Morrison–Woodbury):

$$\text{(BFGS)} \quad B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k}. \quad (6.19)$$

Duality

DFP $\leftrightarrow$ BFGS under $s \leftrightarrow y$, $B \leftrightarrow H$. Compare (6.13) with (6.17), and (6.15) with (6.19). Not a coincidence — it is how they were derived.

Algorithm 6.1 (BFGS)

Given x_0 , tolerance $\epsilon > 0$, inverse Hessian approximation H_0 ; $k \leftarrow 0$;
while $\|\nabla f_k\| > \epsilon$:
 $p_k = -H_k \nabla f_k$;
 $x_{k+1} = x_k + \alpha_k p_k$, α_k from a line search satisfying the Wolfe conditions (3.6);
 $s_k = x_{k+1} - x_k$, $y_k = \nabla f_{k+1} - \nabla f_k$;
 H_{k+1} from (6.17); $k \leftarrow k + 1$.(6.18)

- ▶ Cost per iteration: $O(n^2)$ — matrix–vector products and rank-two updates. No $O(n^3)$ solves or matrix–matrix products.
- ▶ Superlinear rate; robust. Newton is quadratic, but each Newton step costs second derivatives and a linear solve.
- ▶ H_0 : no magic. Finite-difference Hessian inverse at x_0 if affordable; otherwise I or βI with β reflecting the variable scaling. (Better: next slides.)
- ▶ Working with B_k via (6.19) naively costs $O(n^3)$ per step for $B_k p_k = -\nabla f_k$; updating a Cholesky factor of B_k brings it back to $O(n^2)$, with no real advantage in practice.

What it looks like: Rosenbrock from $(-1.2, 1)$

Wolfe line search in all three. Iterations to reduce $\|\nabla f\|$ below 10^{-5} :

Steepest descent	5264
BFGS	34
Newton (inexact)	21

Last four values of $\|x_k - x^*\|$:

steepest descent	BFGS	Newton
1.827×10^{-4}	1.70×10^{-3}	3.48×10^{-2}
1.826×10^{-4}	1.17×10^{-3}	1.44×10^{-2}
1.824×10^{-4}	1.34×10^{-4}	1.82×10^{-4}
1.823×10^{-4}	1.01×10^{-6}	1.17×10^{-8}

Read the columns: SD's error barely moves (linear, $r \approx 1$); BFGS drops by growing factors (superlinear); Newton's exponent doubles (quadratic). Three rates, visible to the eye.

Positive definiteness is preserved (and why that matters)

Problem (6.16) never demanded $H_{k+1} \succ 0$. Yet it holds automatically:

Claim: $H_k \succ 0$ and $y_k^T s_k > 0 \Rightarrow H_{k+1} \succ 0$

For any $z \neq 0$, with $w = z - \rho_k y_k (s_k^T z)$,

$$z^T H_{k+1} z = w^T H_k w + \rho_k (s_k^T z)^2 \geq 0.$$

Zero only if $s_k^T z = 0$ — but then $w = z \neq 0$ and the first term is positive. $\square$

- ▶ So $p_k = -H_k \nabla f_k$ is *always* a descent direction — provided the line search keeps delivering $y_k^T s_k > 0$. The Wolfe conditions, the update, and descent form one closed loop.
- ▶ Invariance under transformations of the variables comes from the choice of W in the norm; other weightings give other updates, none consistently better than BFGS.

Self-correction: the property that separates BFGS from DFP

Can the update go wrong? If $y_k^T s_k$ is tiny but positive, ρ_k is huge and H_{k+1} acquires very large entries. Do rounding errors accumulate until the approximation is garbage?

What is known

- ▶ BFGS has strong **self-correcting** properties: if H_k misjudges the curvature and the iteration slows down as a result, the approximation *repairs itself within a few steps*.
- ▶ DFP is much weaker at this. That is believed to be the reason for its poorer practical performance.
- ▶ Self-correction holds **only with an adequate line search**: the Wolfe conditions sample gradients at points that let the model (6.1) capture the right curvature.

This is a structural fact about the update formula, not a tuning issue. It is why the method tolerates a crude H_0 and an inaccurate line search.

Implementation details that matter

- ▶ **Line search:** Wolfe or strong Wolfe; **always try $\alpha = 1$ first** (Theorem 3.6: it will eventually be accepted, giving the superlinear rate). An *inaccurate* search is cheaper in function evaluations: $c_1 = 10^{-4}$, $c_2 = 0.9$.
- ▶ **Initial matrix:** $H_0 = \beta I$ with β unknown. Too large $\Rightarrow p_0$ too long, many evaluations wasted. Some codes ask for a first-step norm δ and set $H_0 = \delta \|g_0\|^{-1} I$. A better heuristic — after the first step, before the first update:

$$H_0 \leftarrow \frac{y_k^T s_k}{y_k^T y_k} I. \quad (6.20)$$

With $z_k = \bar{G}_k^{1/2} s_k$: $\frac{y_k^T s_k}{y_k^T y_k} = \frac{z_k^T z_k}{z_k^T \bar{G}_k z_k}$ (6.21), a Rayleigh quotient of $\bar{G}_k^{-1}$ — so (6.20) sizes H_0 like an eigenvalue of $\nabla^2 f_0^{-1}$.

- ▶ **Do not skip updates.** If the line search is Armijo backtracking from $\alpha = 1$, $y_k^T s_k > 0$ is *not* guaranteed (a step longer than 1 may be needed). Some codes set $H_{k+1} = H_k$ when it fails. **Not recommended:** skips happen too often and the model loses curvature. The fix is the *damped* BFGS update of Chapter 18.

Symmetric rank-one: the simplest update that can satisfy the secant equation

Try $B_{k+1} = B_k + \sigma v v^T$, $\sigma = \pm 1$. The secant equation $B_{k+1} s_k = y_k$ becomes

$$y_k = B_k s_k + \sigma (v^T s_k) v, \quad (6.22)$$

so v must be a multiple of $y_k - B_k s_k$; substituting $v = \delta(y_k - B_k s_k)$ fixes σ, δ . The *only* symmetric rank-one update satisfying the secant equation:

$$\text{(SR1)} \quad B_{k+1} = B_k + \frac{(y_k - B_k s_k)(y_k - B_k s_k)^T}{(y_k - B_k s_k)^T s_k}, \quad H_{k+1} = H_k + \frac{(s_k - H_k y_k)(s_k - H_k y_k)^T}{(s_k - H_k y_k)^T y_k}. \quad (6.24-6.25)$$

So simple it has been rediscovered repeatedly.

Two features, opposite in sign

- ▶ B_{k+1} need **not** be positive definite even if B_k is. A major drawback in the line-search era; with trust regions, *an advantage* — indefinite approximations can reflect indefinite Hessians.
- ▶ The denominator can **vanish**. Even for a convex quadratic, some steps admit no symmetric rank-one secant update at all.

The three cases, and the skipping safeguard

- 1 $(y_k - B_k s_k)^T s_k \neq 0$: unique update, (6.24).
- 2 $y_k = B_k s_k$: secant equation already holds; the only rank-one solution is $B_{k+1} = B_k$.
- 3 $y_k \neq B_k s_k$ but $(y_k - B_k s_k)^T s_k = 0$: **no symmetric rank-one solution exists.**

Case 3 says rank one is not enough freedom in general — the argument that leads back to rank-two BFGS, where positive definiteness (hence nonsingularity) is guaranteed.

Safeguard (used by most implementations)

Apply (6.24) only if

$$|s_k^T (y_k - B_k s_k)| \geq r \|s_k\| \|y_k - B_k s_k\|, \quad r \approx 10^{-8}; \quad (6.26)$$

otherwise $B_{k+1} = B_k$.

Why skipping is fine here but was discouraged for BFGS

$s_k^T (y_k - B_k s_k) \approx 0$ requires a specific alignment of vectors — *rare*. And it means $s_k^T \tilde{G}_k s_k \approx s_k^T B_k s_k$: the curvature of B_k along s_k is *already right*, so there is nothing to learn. The BFGS condition $s_k^T y_k > 0$, by contrast, fails easily whenever the line search is too short — skipping there is frequent and harmful.

Why SR1 at all

- 1 The safeguard (6.26) appears to prevent breakdown and instability in practice.
- 2 SR1 matrices tend to be **good approximations of the true Hessian** — often better than BFGS (Theorems 6.1, 6.2 below make this precise).
- 3 In **constrained** problems (Ch. 18) and for **partially separable** functions (Ch. 7) the curvature condition $y_k^T s_k > 0$ *cannot be imposed*, so BFGS is not recommended — and there an indefinite approximation is exactly what one wants.

The natural home of SR1 is a trust region

Line search needs $B_k \succ 0$ for descent; a trust region (Ch. 4) minimizes the model inside a ball and does not care. Theorem 4.1 handles indefinite B_k by construction.

Algorithm 6.2 (SR1 Trust-Region Method)

```
Given  $x_0, B_0, \Delta_0$ , tolerance  $\epsilon, \eta \in (0, 10^{-3}), r \in (0, 1); k \leftarrow 0$ ;  
while  $\|\nabla f_k\| > \epsilon$ :  
    solve  $\min_s \nabla f_k^T s + \frac{1}{2} s^T B_k s$  s.t.  $\|s\| \leq \Delta_k$  for  $s_k$ ; (6.27)  
     $y_k = \nabla f(x_k + s_k) - \nabla f_k$ ;    $\text{ared} = f_k - f(x_k + s_k)$ ;    $\text{pred} = -(\nabla f_k^T s_k + \frac{1}{2} s_k^T B_k s_k)$ ;  
    if  $\text{ared}/\text{pred} > \eta$ :  $x_{k+1} = x_k + s_k$  else  $x_{k+1} = x_k$ ;  
    radius:  $\text{ared}/\text{pred} > 0.75$  and  $\|s_k\| > 0.8\Delta_k \Rightarrow \Delta_{k+1} = 2\Delta_k$ ;    $< 0.1 \Rightarrow \Delta_{k+1} = \frac{1}{2}\Delta_k$ ;   else  
    unchanged;  
    if (6.26) holds:  $B_{k+1}$  by (6.24) even if  $x_{k+1} = x_k$ ; else  $B_{k+1} = B_k$ ;  
     $k \leftarrow k + 1$ .
```

Algorithm 4.1 with SR1 supplying B_k ; the radius rule is one reasonable choice among many.

Update on failed steps too

A rejected step means B_k was wrong along s_k . If it is not corrected, similar steps will be generated and rejected again — and superlinear convergence is lost. So the update uses y_k from the trial point whether or not the step was accepted.

How good are SR1 approximations? (Theorems 6.1, 6.2)

Theorem 6.1 (quadratic $f = b^T x + \frac{1}{2}x^T A x$, $A \succ 0$, unit steps)

For any x_0 and any symmetric H_0 , SR1 (6.25) reaches the minimizer in at most n steps, provided the denominators are nonzero. If n steps are taken with linearly independent directions, then $H_n = A^{-1}$.

Key step: the secant equation holds along *all* previous directions, $H_k y_j = s_j$ for $j < k$ (6.29) — by induction, because $(s_k - H_k y_k)^T y_j = s_k^T y_j - y_k^T s_j = 0$ when $y = A s$. Then $H_n A s_j = s_j$ for a basis, so $H_n = A^{-1}$. **No assumption on the line search.** BFGS gets this only with exact line searches (Thm 6.4).

Theorem 6.2 (general f)

Let $\nabla^2 f$ be bounded and Lipschitz near x^* , $x_k \rightarrow x^*$, (6.26) hold for all k , and the steps be *uniformly linearly independent*. Then $\|B_k - \nabla^2 f(x^*)\| \rightarrow 0$.

“Uniformly linearly independent”: the steps do not collapse into a lower-dimensional subspace. Usually true in practice, not always.

Contrast with BFGS, which is superlinear *without* $B_k \rightarrow \nabla^2 f(x^*)$ (Dennis–Moré). SR1 actually converges to the Hessian; that is why its matrices are useful as *Hessians*, not only as direction generators.

One family containing BFGS, DFP, and SR1

$$B_{k+1} = B_k - \frac{B_k s_k s_k^T B_k}{s_k^T B_k s_k} + \frac{y_k y_k^T}{y_k^T s_k} + \phi_k (s_k^T B_k s_k) v_k v_k^T, \quad v_k = \frac{y_k}{y_k^T s_k} - \frac{B_k s_k}{s_k^T B_k s_k}. \quad (6.32-6.33)$$

$\phi_k = 0$: BFGS. $\phi_k = 1$: DFP. Equivalently $B_{k+1} = (1 - \phi_k) B_{k+1}^{\text{BFGS}} + \phi_k B_{k+1}^{\text{DFP}}$.

- ▶ Every member satisfies the secant equation (both endpoints do).
- ▶ $\phi_k \in [0, 1]$ — the **restricted Broyden class** — preserves positive definiteness when $s_k^T y_k > 0$.
- ▶ SR1 is in the family: $\phi_k = \frac{s_k^T y_k}{s_k^T y_k - s_k^T B_k s_k}$ — generally *outside* $[0, 1]$.

Where does positive definiteness break? The last term is rank one; by eigenvalue interlacing it raises eigenvalues for $\phi_k > 0$, lowers them for $\phi_k < 0$. B_{k+1} becomes singular at

$$\phi_k^c = \frac{1}{1 - \mu_k}, \quad \mu_k = \frac{(y_k^T B_k^{-1} y_k)(s_k^T B_k s_k)}{(y_k^T s_k)^2} \geq 1 \text{ (Cauchy-Schwarz)}, \quad (6.37-6.38)$$

so $\phi_k^c \leq 0$: all $\phi_k > \phi_k^c$ keep $B_k \succ 0$. Some negative ϕ is allowed.

What the restricted class does on a quadratic (Theorems 6.3, 6.4)

Theorem 6.3 (any line search)

On $f = b^T x + \frac{1}{2} x^T A x$ with $\phi_k \in [0, 1]$, the eigenvalues $\lambda_1^k \leq \dots \leq \lambda_n^k$ of $A^{1/2} B_k^{-1} A^{1/2}$ satisfy

$$\min\{\lambda_i^k, 1\} \leq \lambda_i^{k+1} \leq \max\{\lambda_i^k, 1\}. \quad (6.36)$$

This fails for $\phi_k \notin [0, 1]$.

All eigenvalues equal to 1 means $B_k = A$. So (6.36) says: each eigenvalue moves *monotonically toward* 1 — never past it, never away. Holds with *inexact* line searches. (Yet controlled negative ϕ_k can outperform BFGS; the picture is not settled.)

Theorem 6.4 (exact line search, $\phi_k \geq \phi_k^c$)

(i) The iterates are *independent of* ϕ_k and converge in at most n steps. (ii) $B_k s_j = y_j$ for all $j < k$. (iii) **With $B_0 = I$ the iterates coincide with conjugate gradient**; the directions are A -conjugate. (iv) $B_n = A$.

With $B_0 \neq I$: identical to *preconditioned* CG with preconditioner B_0 . Mainly theoretical — inexact line searches make the family members behave very differently — but this analysis steered the development of the whole field.

What can and cannot be proved

BFGS and SR1 are remarkably robust in practice. Yet:

It is *not known* whether BFGS converges to a stationary point for general nonlinear f from an arbitrary x_0 and B_0 . Global results assume convexity (or properties of the iterates). Local superlinear results, by contrast, hold under reasonable assumptions.

The difficulty: B_k evolves by updates, so its condition number cannot be bounded the way §3.2 needed. New tools are required — the **trace** (sum of eigenvalues) and the **determinant** (product), combined into

$$\psi(B) = \text{trace}(B) - \ln \det(B) > 0. \quad (6.49)$$

Notation for this section: $G(x) = \nabla^2 f(x)$, Euclidean norms.

Global convergence of BFGS on convex functions (Theorem 6.5)

Assumption 6.1

$f \in C^2$; the level set $\mathcal{L} = \{x : f(x) \leq f(x_0)\}$ is convex and $m\|z\|^2 \leq z^T G(x)z \leq M\|z\|^2$ on $\mathcal{L}$. (6.39)

Theorem 6.5

For any $B_0 \succ 0$ and any x_0 satisfying Assumption 6.1, Algorithm 6.1 with $\epsilon = 0$ converges to the unique minimizer x^* .

Shape of the proof. Set $m_k = y_k^T s_k / s_k^T s_k \geq m$ and $M_k = y_k^T y_k / y_k^T s_k \leq M$ (6.42–6.43). The update (6.19) changes trace and determinant by known amounts,

$$\text{trace}(B_{k+1}) = \text{trace}(B_k) - \frac{\|B_k s_k\|^2}{s_k^T B_k s_k} + \frac{\|y_k\|^2}{y_k^T s_k}, \quad \det(B_{k+1}) = \det(B_k) \frac{y_k^T s_k}{s_k^T B_k s_k}, \quad (6.44\text{--}6.45)$$

and with $\cos \theta_k = s_k^T B_k s_k / (\|s_k\| \|B_k s_k\|)$ one gets

$$0 < \psi(B_{k+1}) \leq \psi(B_0) + c(k+1) + \sum_{j=0}^k \ln \cos^2 \theta_j. \quad (6.51)$$

θ_k is exactly the angle between $-\nabla f_k$ and p_k of Chapter 3. If $\cos \theta_j \rightarrow 0$ the right side goes to $-\infty$, contradicting $\psi > 0$. So a subsequence has $\cos \theta_j \geq \delta$, Zoutendijk gives $\liminf \|\nabla f_k\| = 0$, and strong convexity finishes.

Extends to $\phi_k \in [0, 1)$, not to DFP; refinement: $\sum_k \|x_k - x^*\| < \infty$ (6.52).

Superlinear convergence of BFGS (Theorem 6.6)

Assumption 6.2

G is Lipschitz at x^* : $\|G(x) - G(x^*)\| \leq L\|x - x^*\|$ near x^* .

Theorem 6.6 — general (nonconvex) f

If the BFGS iterates converge to a minimizer x^* satisfying Assumption 6.2, and $\sum_k \|x_k - x^*\| < \infty$ (6.52), then the convergence is superlinear.

Shape of the proof. Work in scaled variables $\tilde{s}_k = G_*^{1/2} s_k$, $\tilde{y}_k = G_*^{-1/2} y_k$, $\tilde{B}_k = G_*^{-1/2} B_k G_*^{-1/2}$; the update keeps the BFGS form, so the ψ -inequality (6.53) applies again. Lipschitz continuity gives $\|\tilde{y}_k - \tilde{s}_k\| \leq \bar{c} \epsilon_k \|\tilde{s}_k\|$ with $\epsilon_k = \max\{\|x_{k+1} - x^*\|, \|x_k - x^*\|\}$, and (6.52) makes $\sum \epsilon_k < \infty$. Summing (6.53) then forces both $\ln \cos^2 \tilde{\theta}_k \rightarrow 0$ and $\tilde{q}_k \rightarrow 1$, which together are

$$\lim_{k \rightarrow \infty} \frac{\|(B_k - G_*)s_k\|}{\|s_k\|} = 0$$

— the Dennis–Moré condition (3.36). Theorem 3.6: $\alpha_k = 1$ is accepted, rate superlinear.

Note what was not needed: $B_k \rightarrow G_*$. Accuracy along the step directions suffices. That is the “surprising and delightful” fact of Chapter 3, now proved for BFGS.

Convergence of SR1 (Theorem 6.7)

Less is known. No global result like Theorem 6.5, no local result like Theorem 6.6 — except on quadratics. But for the trust-region version:

Theorem 6.7

Let Algorithm 6.2 generate $\{x_k\}$ and suppose (c1) the iterates stay in a closed bounded convex set D on which $f \in C^2$ has a unique stationary point x^* ; (c2) $\nabla^2 f(x^*) \succ 0$, $\nabla^2 f$ Lipschitz near x^* ; (c3) $\{B_k\}$ is bounded; (c4) (6.26) holds at every iteration — the update is never skipped. Then $x_k \rightarrow x^*$ and

$$\lim_{k \rightarrow \infty} \frac{\|x_{k+n+1} - x^*\|}{\|x_k - x^*\|} = 0 \quad \text{— } (n+1)\text{-step superlinear convergence.}$$

- ▶ No exact subproblem solution required.
- ▶ (c3) is needed for SR1, not for BFGS.
- ▶ B_k may be indefinite at any iteration, so the trust-region bound may stay active forever. Yet under these assumptions the fraction of k with $B_k \succeq 0$ tends to 1 — regardless of B_0 .

Chapter 6 in one table

	BFGS	DFP	SR1
Rank of update	2	2	1
Derived from	closest H (6.16)	closest B (6.9)	uniqueness (6.22)
Preserves $\succ 0$	yes, if $s^T y > 0$	yes, if $s^T y > 0$	no
Needs $s^T y > 0$	yes (Wolfe)	yes	no
Can break down	no	no	yes (denominator); skip (6.26)
Self-correcting	strong	weak	—
Natural framework	line search	line search	trust region
$B_k \rightarrow \nabla^2 f(x^*)$?	not needed (D–M)	—	yes, Thm 6.2
Global convergence	convex f (Thm 6.5)	open	Thm 6.7 (c1–c4)
Local rate	superlinear (Thm 6.6)	superlinear	$(n+1)$ -step superlinear
Broyden ϕ	0	1	$s^T y / (s^T y - s^T B s)$

Three ideas to carry forward

The secant equation is Newton's method learning curvature from gradient differences. Wolfe's curvature condition is what makes BFGS well defined; it was never just a line-search nicety. Superlinear convergence needs accuracy along the steps, not convergence of the matrix.

Homework

Derivations

- 1 (Ex. 6.1) f strongly convex $\Rightarrow s_k^T y_k > 0$ for any x_k, x_{k+1} .
- 2 Derive the B -form (6.19) from the H -form (6.17) via Sherman–Morrison–Woodbury.
- 3 Expand $z^T H_{k+1} z$ from (6.17) and confirm $z^T H_{k+1} z = w^T H_k w + \rho_k (s_k^T z)^2$, $w = z - \rho_k y_k (s_k^T z)$.
- 4 Complete the SR1 derivation from (6.22): find σ, δ ; show uniqueness.
- 5 (Ex. 6.8–6.11) Prove (6.44), (6.45); show $1 - t + \ln t \leq 0$ and $\psi(B) > 0$ for $B \succ 0$.

Implementation

6. Algorithm 6.1 with your Ch. 3 strong-Wolfe search ($c_1 = 10^{-4}$, $c_2 = 0.9$), H_0 scaled by (6.20). Rosenbrock from $(-1.2, 1)$: print the last five $\|x_k - x^*\|$ and confirm superlinear decay.
7. Replace Wolfe by Armijo backtracking from $\alpha = 1$. Log the sign of $y_k^T s_k$ each iteration and what happens to H_k when it goes negative — with and without skipping.
8. Algorithm 6.2 with dogleg or CG–Steihaug from Ch. 4, $B_0 = I$. Log skipped updates and $\|B_k - \nabla^2 f(x_k)\|$; compare with BFGS. Then check Theorem 6.4(iii): BFGS, $B_0 = I$, exact line search on a 3×3 quadratic reproduces the CG iterates.

Next: Ch. 7 — *large scale: L-BFGS, inexact Newton, and what to do when n is 10^6 .*